ICO Std 2001:2026

DPP-CQ White Paper v2.0

Digital Product Passport for Cultural & Quality Goods

Standard Architecture and Core Specifications

Version 2.0.0-draft — Public Review Draft (2nd round, Sep 1 – Oct 15, 2026)

International Communication Organization (ICO)

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DPP-CQ: Digital Product Passport for Cultural & Quality Goods Standard Architecture and Core Specifications

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DPP-CQ: Digital Product Passport for Cultural & Quality Goods

Standard Architecture and Core Specifications

(ICO Digital Trust Standards Framework v2.0.0-draft)

Abstract

Against the backdrop of deep integration between the digital economy and global trade, cross-border product circulation is evolving from *physical flow* to *trusted flow*. Digital Product Passports (DPPs), as an emerging digital trust infrastructure, have achieved significant progress in the environmental data domain for industrial products, yet notable gaps remain in their coverage of cultural products, specialty agricultural goods, geographical indication products, and other categories.

This document systematically presents the overall architecture, core specifications, governance mechanisms, and implementation roadmap of the **DPP-CQ Digital Product Passport for Cultural & Quality Goods standards framework**. Built upon mature technologies such as W3C DIDs, Verifiable Credentials (VCs), and standards-based selective disclosure (SD-JWT / BBS+), this framework focuses on establishing digital trust in the cultural value and quality dimensions, addressing the inclusivity gap in existing DPP standards ecosystems.

The core standard, **ICO Std 2001 (DPP-CQ)**, is a digital passport specification for cultural and quality products. It supports dual carriers of QR codes and NFC, provides a three-tier verification mechanism, and protects sensitive craftsmanship and commercial data through selective-disclosure proofs (SD-JWT / BBS+). The standards framework employs a multi-stakeholder governance model to ensure openness, neutrality, and inclusivity.

This document is issued as a Public Review Draft. We sincerely invite governments, enterprises, academic institutions, and non-governmental organizations worldwide to participate in building a more inclusive global digital trust ecosystem.

v2.0.0-draft update (Aug 2026): Following the first public review, this revision substantially expands the standard's international interoperability and compliance posture: a GS1 Digital Link / GTIN interop profile, three-tier physical carrier specification (open QR + SDM NFC + tamper-evident NFC) aligned with EN 18220, two normative credential format profiles (W3C Data Integrity with BBS+ and IETF SD-JWT VC / RFC 9529), an optional Chinese national cryptography suite (SM2/SM3/SM4), an ISO 14067-aligned sustainability data module (quality + sustainability dual-dimension model), AI-assisted assessment transparency provisions aligned with GB/T 47507-2026 and EU AI Act Art. 50, data lifecycle governance metadata, and UNTP-style conformity claims. All v2.0 additions are backward compatible with v1.x credentials.

Keywords: Digital Product Passport; Cross-Cultural Digital Trust; Quality Assets; DPP-CQ; Decentralized Identifiers; Verifiable Credentials; Selective Disclosure (SD-JWT / BBS+); GS1 Digital Link; Multi-Stakeholder Governance

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Scope

This document describes the **overall architecture, core specifications, governance mechanisms, and implementation roadmap** of the **DPP-CQ Digital Product Passport for Cultural & Quality Goods standards framework**, with particular focus on the technical framework and application model of the core product-level standard **ICO Std 2001-2026 (DPP-CQ)**.

Applicability: - Categories with cultural value and quality attributes, including cultural products, Intangible Cultural Heritage-related products, Geographical Indication (GI) products, specialty agricultural goods, and haute couture products **-** Application scenarios including product identity verification, quality grading, traceability verification, and digital confirmation of cultural value **-** Technical integration and system interoperability for all participating parties, including producers, certification bodies, trading platforms, and consumers

Exclusions: - Highly regulated special categories such as pharmaceuticals and medical devices (which must comply with sector-specific regulatory requirements) **-** High-sensitivity scenarios at the financial-grade or personal safety level, such as financial payment and identity authentication **-** Replacement of mandatory compliance requirements such as statutory product safety and quality inspection in respective jurisdictions

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Purpose

This document aims to: 1. Systematically present the design philosophy and overall architecture of the ICO standards framework, providing a panoramic understanding for all stakeholders; 2. Define the core technical specifications and interoperability requirements of the DPP-CQ standard, guiding technical implementation; 3. Describe the governance mechanisms and participation pathways of the standards framework, inviting global stakeholders to co-build; 4. Solicit feedback from all sectors of society to promote continuous improvement and widespread adoption of the standards.

Status of This Document

This document is a **Public Review Draft** published by the International Communication Organization (ICO).

The content of this document may be revised based on community feedback. The official version shall be as officially published by ICO. This v2.0.0-draft incorporates the results of the first public review (July 6 – August 20, 2026) and opens a **second review period: September 1 – October 15, 2026**, focused on the new international interoperability and compliance annexes. Feedback may be submitted through the following channels:

- GitHub Issues: https://github.com/ICO-cloud/dpp-cq-standard/issues
- Email: info@icoun.org

This document does not constitute a legally binding instrument, nor does it represent the official position of ICO members. Formal publication of the standard requires approval by the ICO Standards Council.

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For the complete list of contributors and detailed revision history, please refer to the commit history of the GitHub repository.

Revision History

Version	Date	Description of Changes
v2.0.0-draft	Aug 2026	International interoperability & compliance expansion: GS1 Digital Link/GTIN interop profile; three-tier carrier specification (EN 18220 aligned); SD-JWT VC (RFC 9529) profile alongside BBS+; optional SM2/SM3/SM4 national cryptography suite; ISO 14067 sustainability data module (quality+sustainability dual-dimension); AI-assisted assessment transparency (GB/T 47507-2026, EU AI Act Art. 50); data lifecycle governance metadata; UNTP-style conformity claims; conformity declarations and PIA summary published; all additions backward compatible with v1.x
v1.3	July 2026	Format standardization: aligned with international technical specification formats; added abstract, document status, copyright notice, and revision history; annexes classified as normative/informative; unified section numbering; main title updated to DPP-CQ; verified accuracy of data citations
v1.2	July 2026	Added compliance and legal framework chapter, DPP-CQ technical specification annex, intellectual property policy, standards version management strategy, and FAQ; strengthened references to UNESCO/WIPO/UNECE
v1.1	July 2026	Framework restructuring and refinement: optimized architecture description; added risks and challenges chapter, pilot framework, and UNECE interoperability initiative references
v1.0	July 2026	First public release (Public Review Draft)
v0.9	July 2026	Internal working draft

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1. Introduction

In an era of deep integration between the digital economy and global trade, cross-border product circulation is transitioning from *physical flow* to *trusted flow*. However, significant **inclusivity gaps** exist in the current global digital product verification ecosystem:

- Existing Digital Product Passport (DPP) standards primarily focus on environmental and circular economy data for industrial products, and do not adequately cover categories such as cultural products, specialty agricultural goods, and geographical indication products.
- A vast number of products with unique cultural value and quality advantages lack digital trust-transmission tools, making it difficult for them to fully realize their value in global markets. According to a joint OECD/EUIPO study, global trade in counterfeit goods reached USD 467 billion in 2021, accounting for 2.3% of global imports. Among these, geographical indication products and specialty cultural products, with their high recognizability and premium pricing potential, have become prime targets for counterfeiting. Specialty products from developing countries are severely undervalued due to lack of trust.
- Developing economies face multiple challenges in digital trust infrastructure development, including technological, standardization, and capacity gaps. The UN *Global Digital Compact* (GDC) explicitly calls for increased support for digital capacity building in developing countries to ensure that digital technologies benefit all people.

The International Communication Organization (ICO), as a multi-stakeholder international cooperation platform, is committed to advancing **Global Digital Inclusion** and **Cross-Cultural Digital Trust**. This document presents the overall architecture, core standards, governance mechanisms, and implementation roadmap of the ICO standards framework, and invites global stakeholders to join us in building a more inclusive global digital trust ecosystem.

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2. Mission & Vision

2.1 Mission

To empower the digital preservation and value transmission of global cultural and quality assets through open standards.

2.2 Vision

To build a **globally applicable, technology-neutral, and culturally inclusive** digital trust standards framework, ensuring that every product carrying cultural value and quality commitment possesses a verifiable, traceable, and mutually recognizable digital identity. By 2030, achieve 100,000+ cultural and quality products with globally mutually recognizable digital identities, driving global trade from *price competition* to *value competition*.

2.3 Core Principles

Principle	Description	Practical Manifestation
Open	Standards are open to all stakeholders, regardless of country, region, or ownership structure	Free adoption with no membership fee barriers
Neutral	Technology-agnostic architecture, not tied to any specific vendor or technology stack	Supports multiple blockchain and cloud platforms
Inclusive	Fully considers the needs of different cultural backgrounds and development stages	Supports lightweight implementations for low-resource environments
Interoperable	Based on international technical standards, ensuring connectivity with existing and future systems	Compatible with W3C, ISO, GS1, UNECE, and other standards
Trustworthy	Verifiable data, traceable processes, and adjudicatable disputes	Three-tier verification + Root Resolver Network + selective-disclosure privacy protection
Privacy by Design	Privacy protection embedded in architectural design; data minimization	Selective disclosure (SD-JWT / BBS+)

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3. Background & Rationale

3.1 The Trust Deficit in Global Trade

In current cross-border global trade, categories such as specialty agricultural goods, cultural products, and handicrafts face pervasive trust challenges: consumers struggle to verify the true origin, quality grade, and cultural background of products; purchasers lack efficient and reliable supply chain traceability tools; authentication of geographical indication products is costly; and cultural value and craft heritage cannot be effectively transmitted through traditional channels.

A direct consequence of this trust deficit is the "bad money drives out good money" effect — genuinely superior products and producers who uphold traditional craftsmanship are unable to obtain returns commensurate with their value in the global market.

3.2 Coverage Gaps in Existing Standards

Existing DPP standards are primarily led by advanced economies, focusing on environmental footprints and circular economy data for industrial products, compliance-driven mandatory information disclosure, and serving the market access requirements of developed economies. Significant standard gaps remain in the following areas:

Gap Area	Current State	ICO Positioning
Digital confirmation of cultural product rights	No unified cultural value data standard	Establish cultural data models and verification specifications
Cross-border mutual recognition of Geographical Indications	WIPO has legal framework but no technical standards	Develop GI digital mutual recognition technical specifications
Quality grading for specialty agricultural goods	Individual national standards, no global unified digital framework	Establish cross-cultural quality assessment methodology
Digital capacity in developing economies	High technical barriers, high costs, low participation	Provide lightweight, low-threshold solutions

3.3 Interoperability with Existing Standards

ICO standards adopt a strategy of **"baseline compatibility, differentiated value"**:

- **W3C DID/VC**: Used as the underlying identity and credential technology foundation, fully compatible; v2.0 supports both Data Integrity proofs (BBS+) and SD-JWT VC (RFC 9529)
- **GS1 Identification System**: GTIN/GLN identifiers and GS1 Digital Link resolution are supported through a normative interop profile; supply-chain events map to EPCIS 2.0/CBV 2.0; DID and GTIN are bound as complementary identifiers
- **EU DPP / ESPR / EN 18219-18223**: Open data carriers (EN 18220), unique identifiers (EN 18219) and sustainability data attributes are aligned at field level; conformity claims use UNTP-style vocabulary
- **UN/CEFACT UNTP**: Conformity claim and interoperability vocabulary adopted as the multilateral bridge for cross-regime data exchange
- **ISO standards**: Compatible with ISO 22000 (food safety), ISO 14067 (product carbon footprint), ISO 26000 (social responsibility), ISO 22739 (blockchain terminology)
- **Chinese national standards**: AI trustworthiness (GB/T 47507-2026) and commercial cryptography (SM2/SM3/SM4) supported for domestic deployments

DPP-CQ does not replace statutory conformity assessment or sectoral certification held by producers; it carries and references such attestations in a verifiable, machine-readable form.

3.4 Technology Maturity Window

Technologies such as W3C DIDs (Decentralized Identifiers), Verifiable Credentials, blockchain-based evidence anchoring, and selective-disclosure credential technology (SD-JWT, BBS+) have entered mature application phases, providing the technical foundation for building a globally applicable open standard for the cultural and quality dimensions.

Meanwhile, international documents and data such as UNESCO's *Operational Directives for the Implementation of the Convention for the Safeguarding of the Intangible Cultural Heritage* (2016/2018 edition) and WIPO's *World Intellectual Property Indicators 2024* all indicate the urgency and necessity of leveraging digital technology to protect cultural assets and geographical indications. The timing is opportune for establishing open standards oriented toward the cultural and quality dimensions.

4. Standards Architecture

4.1 Four-Layer Architecture

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┌──────────────────────────────────────────────────────┐
│ Layer 4  Governance & Compliance                      │
│  Standards development process · Multi-stakeholder    │
│  governance · Dispute resolution · Assessment rules │
│  rules                                                │
├──────────────────────────────────────────────────────┤
│ Layer 3  System & Assessment                         │
│  Institutional credibility assessment · Cross-        │
│  cultural trust framework · AI content provenance     │
├──────────────────────────────────────────────────────┤
│ Layer 2  Product Standards                           │
│  Cultural & Quality DPP · Haute couture · GI mutual   │
│  recognition · ICH craftsmanship                      │
├──────────────────────────────────────────────────────┤
│ Layer 1  Base Protocols                              │
│  Digital identifiers · Data provenance · Credential   │
│  formats · Naming conventions                         │
└──────────────────────────────────────────────────────┘
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4.2 Standards Numbering System

Series	Layer	Positioning	Number Range
1000 Series	Base Protocols Layer	Underlying technical specifications, common foundation for all upper-layer standards	1001–1099
2000 Series	Product Standards Layer	Specific data standards and verification specifications for different product categories	2001–2099
3000 Series	System & Assessment Layer	Standards at the institutional assessment, trust system, and methodology level	3001–3099
4000 Series	Governance & Compliance Layer	Standards development processes, governance rules, dispute resolution mechanisms	4001–4099

4.3 Initial Core Standards Inventory

Standard Number	Standard Name	Status	Expected Publication
ICO Std 1001-2026	Global Information Integrity & Data Provenance	Draft	2026 Q3
ICO Std 1002-2026	Digital Credential Format Specification	Under Development	2026 Q3
ICO Std 2001-2026	**Digital Product Passport for Cultural & Quality Goods (DPP-CQ)**	**Public Review Draft**	**2026 Q3**
ICO Std 2002-2026	Global Haute Couture & Cultural Attire Quality Standard	Planned	2026 Q4
ICO Std 2003-2026	Global Framework for GI & Origin Digital Mutual Recognition	Planned	2027 Q1
ICO Std 3001-2026	Global Digital Credibility Assessment (GDCC)	Draft	2026 Q4
ICO Std 4001-2026	Standards Development & Publication Process	Under Development	2026 Q3
ICO Std 4002-2026	Multi-Stakeholder Governance Charter	Under Development	2026 Q3

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5. Core Standards

5.1 ICO Std 2001: Digital Product Passport for Cultural & Quality Goods (DPP-CQ)

Positioning: A digital identity standard for cultural products, specialty agricultural goods, geographical indication products, and ICH handicrafts, providing a unique, trustworthy, and verifiable digital identity credential for every product carrying cultural value and quality commitment.

Core Data Dimensions:

Data Module	Core Fields	Required/Optional
Base Identity	Product unique identifier (DID), issuer information, validity period, category	Required
Quality Data	Quality grade, test report summary, certification credentials, batch information	Category-required
Cultural Data	Cultural background, craft heritage, ICH level, artisan/transmitter information	Category-required
Geographical Indication	Origin, GI certification number, production region data	Required for GI products
Traceability Data	Full lifecycle event records (production/processing/testing/distribution)	Optional
Credential Data	Verifiable credential metadata, signatures, revocation status	Required

Technical Features:

- Based on W3C DID Core and Verifiable Credentials international standards
- Supports dual-carrier verification via QR code and NFC
- Three-tier verification mechanism: Quick Verification (<1s) / Deep Verification / Judicial-Grade Verification
- Hybrid storage architecture: data stored off-chain, hashes anchored on-chain
- **ZKP-enabled Privacy:** The standard enables manufacturers, ICH transmitters, and distribution entities to prove compliance with specific quality grades and origin regulatory requirements to consumers and regulatory authorities

through selective-disclosure proofs (SD-JWT / BBS+), without disclosing core craft recipes, supply chain node identities, or commercial procurement prices — achieving "data usability without visibility, trustworthiness without information leakage"

- Selective Disclosure: Data holders may choose the scope of disclosure based on the verification scenario

Applicable Categories: Geographical indication agricultural products (tea, grains & oils, fruits, medicinal herbs, spices), Intangible Cultural Heritage products (ceramics, embroidery, lacquerware, woodwork), haute couture and cultural attire, premium food and beverages, handicrafts and cultural creative products.

5.2 ICO Std 1001: Global Information Integrity & Data Provenance

Defines the credibility assessment framework and provenance methodology for digital content and data assets, with particular emphasis on provenance identification specifications for AI-generated content. Covers data integrity verification, provenance chain construction methods, credibility grading assessment, timestamps, and evidence anchoring interfaces.

5.3 ICO Std 3001: Global Digital Credibility Assessment (GDCC)

A digital credibility assessment and evaluation standard for institutions (enterprises, media, NGOs, etc.), constructing an assessment system across multiple dimensions including information transparency, data credibility, governance compliance, and cultural responsibility, with support for cross-cultural scenario adaptation.

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6. Governance

6.1 Multi-Stakeholder Governance Model

The ICO standards framework employs a multi-stakeholder governance model to ensure the openness, representativeness, and credibility of standards. Important decisions follow a consensus-first principle; where consensus cannot be reached, decisions are adopted by a two-thirds majority.

Governance design references best practices from international standards bodies such as W3C and IETF, ensuring: - Balanced representation of economies at different development stages - Diverse participation from government, industry, academia, and civil society - Effective separation of technical decision-making from commercial interests

6.2 Governance Structure

General Assembly (supreme decision-making body): Convened annually to deliberate on strategic direction, budget, and Council elections.

Standards Council (decision-making body): Responsible for strategic decisions, standard approval, budget approval, and final dispute adjudication. Seats are allocated by category: Government Representatives 25% / Industry 25% / Academia 20% / Civil Society 20% / Technical Secretariat 10% (non-voting).

Specialized Committees: - Technical Committee: Responsible for standard technical review and technical roadmap planning - Strategy Committee: Responsible for strategic planning and external cooperation - Compliance & Dispute Committee: Responsible for compliance review and dispute adjudication

Standards Working Groups: - WG-DPP: Digital Product Passport Working Group - WG-GI: Geographical Indication Mutual Recognition Working Group - WG-DID: Decentralized Identifiers and Credentials Working Group - WG-

GDCC: Institutional Credibility Assessment Working Group - WG-Governance: Governance Rules Working Group

6.3 Technical Secretariat

The Technical Secretariat is the neutral executive body of the standards framework, undertaken by a designated entity authorized by the Standards Council. Its core responsibilities: - Maintain reference implementations and open-source code repositories for the standards - Provide technical support and conformance testing platforms - Operate the root resolution infrastructure and trusted registries - Support the technical work of working groups - Organize technical training and developer community building

Key Principle: The Technical Secretariat does not participate in standard voting, ensuring the separation of technical execution from standards development.

6.4 Infrastructure and Root Resolution Mechanism

To ensure high security and continuous availability of data resolution in global cross-border circulation, the ICO Council authorizes the Technical Secretariat to build and operate:

- **ICO Root Resolver Network**: A globally distributed DID resolution infrastructure providing highly available, low-latency identifier resolution and credential status query services
- **Tianji Credibility Registry (TCR)**: A unified credential hash registration and anchoring database

Operating Mechanism: When any third-party certification body or service provider issues Digital Product Passports under ICO Std 2001/2003, the credential hash fingerprints and identifiers (DIDs) generated must be registered and consensus-anchored at the root nodes, ensuring traceability and judicial-grade authenticity verification across networks and geographical regions worldwide.

Security Assurance: - Multi-region redundant deployment with 99.99% availability target - At least one independent third-party security audit per year, with audit reports made public - Major technical upgrades require review by the Technical Committee and approval by the Standards Council

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7. Technical Approach

7.1 Technology Selection Principles

- **International Standards First**: Prioritize adoption of published international standards (W3C, IETF, ISO)
- **Open Source & Open**: Core technology stack based on open-source licenses
- **Extensible Architecture**: Modular design supporting future standard extensions
- **Privacy by Design**: Data minimization principle, implemented via standards-based selective disclosure (SD-JWT / BBS+)

7.2 Core Technology Stack

Layer	Technical Standard	Reference / Compatible Specification
Decentralized Identifiers	W3C DID Core 1.0	W3C Recommendation (Jul 2022)
Verifiable Credentials	W3C Verifiable Credentials Data Model v2.0	W3C Recommendation
Selective Disclosure	SD-JWT (RFC 9529) / BBS+ signatures	IETF RFC 9529; W3C Data Integrity BBS cryptosuite
Data Semantics	JSON-LD 1.1	W3C Recommendation / IETF BCP 19
Cryptographic Signatures	BBS+ / Ed25519 / ECDSA P-256; SM2+SM3 optional	IETF RFC 8032; GB/T 32918 / GB/T 32905 (national crypto option)
Global Product Identification	GS1 GTIN / GLN / GS1 Digital Link 1.6	GS1 General Specifications; EU DPP EN 18219 alignment
Supply Chain Events	GS1 EPCIS 2.0 / CBV 2.0	GS1 Standard (2022); mapping profile defined in v2.0
Data Anchoring	Cryptographic Hash Anchoring	Blockchain-agnostic, multi-chain support
Physical Verification	QR ISO/IEC 18004 (open carrier); NFC ISO/IEC 14443 with SDM (NTAG 424 DNA class)	EN 18220:2026 alignment; NFC Forum NDEF
Sustainability Data	Product carbon footprint per ISO 14067	Optional sustainability module; ESPR data-attribute direction
Interface Specification	RESTful API / OpenAPI 3.0; Digital Link content negotiation	OpenAPI Initiative; GS1 Digital Link resolver pattern

7.3 Multi-Chain and Multi-Cloud Architecture

ICO standards are not bound to any specific blockchain or cloud platform. Through a unified hash anchoring abstraction layer, multiple underlying storage solutions are supported (public chains, consortium chains, distributed hash tables, trusted cloud storage, etc.). Different jurisdictions may select anchoring solutions compliant with local regulations while maintaining cross-chain verification consistency.

7.4 Open Source Strategy

Component	Open Source License	Description
Standards documentation	CC BY 4.0	Free to use, reproduce, and adapt; attribution required
Base protocol reference implementations	Apache 2.0	Fully open, business-friendly
SDKs and development tools	Apache 2.0	Multi-language SDKs, encouraging ecosystem participation
Contribution mechanism	CLA	Contributor License Agreement ensuring clear intellectual property rights

7.5 Intellectual Property Policy

The ICO standards framework adopts an intellectual property policy of "openness first, reasonable protection":

Standard-Essential Patent (SEP) Policy: - Encourages patent holders to license their essential patents on FRAND (Fair, Reasonable, and Non-Discriminatory) terms - Core standards (DIDs, credential formats, and other base layers) prioritize open-source technology solutions free of patent encumbrances - Patent searches and disclosures are conducted prior to standard publication to minimize potential patent risks - An Intellectual Property Working Group is established to oversee patent policy development and dispute resolution

Trademark and Logo Policy: - "ICO Verified" mark, "DPP-CQ" mark, and others are registered trademarks of ICO - Products and services meeting DPP-CQ conformance requirements may use official marks subject to compliance requirements; DPP-CQ conformance is an assessment/verification, not statutory certification - Logo usage specifications and supervision mechanisms are established to prevent abuse and misleading - Logo usage by regional nodes and partners requires authorization and review

Copyright Policy: - Copyright in standard texts is held by ICO and made available under the CC BY 4.0 license - Copyright in reference implementations and tool code is held by contributors and open-sourced under the Apache 2.0 license - Third-party derivative works must comply with the corresponding open-source licenses and indicate source

7.6 Standards Version Management

Version Numbering Scheme: Uses the format "major.minor.patch" (e.g., v1.2.3) - **Major version**: Architectural expansion of the standard's scope (e.g. v2.0 adds the GS1 interop profile, SD-JWT VC format and sustainability module); all v2.0 data additions are OPTIONAL so v1.x credentials remain valid - **Minor version**: Backward-compatible functional additions within the same architecture - **Patch version**: Backward-compatible bug fixes

Backward Compatibility Commitments: - Minor version updates maintain backward compatibility of data formats and interfaces - Major version updates provide at least 12 months of transition period and migration tools - The validity of issued credentials is not affected by standard version updates

Standards Lifecycle: - Working Draft (WD) → Committee Draft (CD) → Public Review Draft (PR) → International Standard (IS) - International standards are reviewed every 5 years to ensure continued validity - Obsolete standards initiate revocation procedures with advance notice to all adopters

7.7 Physical Carrier Strategy: Three-Tier, Open by Default

DPP-CQ v2.0 defines three physical carrier levels (normative detail: Carrier Specification):

Level	Carrier	Security profile	Reader requirement
L1	Open QR (GS1 Digital Link in interop profile)	Cloneable; authenticity via registry	Any smartphone camera
L2	SDM secure NFC (NTAG 424 DNA class)	Per-scan cryptographic SUN authentication	Any NFC phone, NDEF open read
L3	Tamper-evident NFC (TagTamper class)	Tamper detection + SDM	Any NFC phone

Two principles govern carrier design:

1. **Open carrier principle (aligned with ESPR Art. 9 and EN 18220):** every product must be readable with generally available means — no proprietary app, registration, or fee. Security is achieved through cryptographic

verification of the resolved data, never through hiding the payload.

2. **Dual carrier for international products:** products in the GS1 interop profile carry both an open QR (universal scanning, retail/customer flows) and an SDM NFC tag (cryptographic authenticity); both resolve to the same passport, and the verification page indicates which carrier was presented.

7.8 Dual-Dimension Data Model: Quality + Sustainability

Existing DPP regimes (notably the EU ESPR framework) organize product data primarily around environmental sustainability and circular economy. DPP-CQ's core differentiation is the **quality and cultural value** dimension. v2.0 formalizes a dual-dimension model rather than choosing between them:

- **Quality & cultural dimension (DPP-CQ core):** quality grade, sensory and laboratory attributes, craftsmanship, ICH/GI status, artisan lineage, assessment methodology and AI-use disclosure.
- **Sustainability dimension (v2.0 optional module):** product carbon footprint per ISO 14067, material composition with recycled content, end-of-life guidance — structured to map onto ESPR delegated-act data attributes as they are adopted per product class.

This design lets a single credential serve both purposes: it satisfies the cultural/quality trust gap DPP-CQ was created to address, while producing machine-readable sustainability data that EU importers, retailers and regulators can consume. The quality assessment itself remains governed by ICO standards (notably Std 3001), preserving the framework's independence.

7.9 AI-Assisted Assessment Transparency

Where AI systems support quality assessment (data extraction, pre-scoring, image recognition), DPP-CQ adopts transparency and human-oversight rules aligned with **GB/T 47507-2026** (AI trustworthiness general rules) and the transparency direction of **EU AI Act Art. 50**:

- AI participation **MUST** be disclosed in the credential (`qualityAttributes.assessment.aiDisclosure`).
- The final grading decision **MUST** be made by a qualified human assessor; the credential identifies the decision role and review process.
- The assessment methodology reference and full report **MUST** be retrievable via the credential (`reportReference`).

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8. Compliance & Legal Framework

8.1 Data Sovereignty and Cross-Border Compliance Architecture

ICO standards employ an architectural design of **"data localization, hash cross-border"**, fundamentally reducing cross-border data compliance risks:

- **Raw data stored locally**: Detailed product quality, craftsmanship, and supply chain data are stored locally by the issuer or certification body, with no mandatory cross-border transmission
- **Hash fingerprint anchoring**: Only hash fingerprints of data (one-way irreversible) are registered and anchored at root nodes; hashes themselves contain no identifiable original information

- **Selective-disclosure verification**: Sensitive data is verified via SD-JWT / BBS+ selective disclosure; verifiers receive only the claims or compliance conclusions they are authorized for, without access to undisclosed raw data

This architecture is compatible with China's *Data Security Law* and *Personal Information Protection Law*, the EU's General Data Protection Regulation (GDPR), and the data sovereignty principles of most countries.

8.2 Compliance Adaptation by Major Jurisdictions

Jurisdiction	Core Compliance Requirements	ICO Standard Adaptation
China	Data classification and grading, security assessment for outbound transfer of important data, critical information infrastructure protection; commercial cryptography requirements; AI trustworthiness (GB/T 47507-2026)	Localized data storage, hash cross-border, independent deployment of domestic root nodes; optional SM2/SM3/SM4 cryptography suite; AI assessment transparency & human-oversight provisions; data lifecycle governance metadata
European Union	GDPR data protection; ESPR digital product passport & sustainability data; EN 18219/18220 series; EU AI Act transparency	SD-JWT/BBS+ selective disclosure, data minimization; GS1 Digital Link & EN 18220-aligned open carriers; ISO 14067 sustainability module; conformity claims; AI-use disclosure
Southeast Asia	Significant variation in national data protection laws, differing localization requirements	Flexible regional node deployment, configurable data storage strategies; lightweight L1 QR carrier for cost sensitivity
Middle East / GCC	Developing data localization and halal/cultural-product traceability regimes; strong interest in digital trade infrastructure	Regional node deployment; cultural-quality framing compatible with heritage-product protection; GS1/UNTP interop for trade
Global South	Relatively weak data regulations, insufficient digital capacity	Lightweight solutions, capacity building support, technology transfer; zero-licensing-fee adoption (see three red lines: no charging for standards use)

8.2b International Interoperability Position

DPP-CQ is an **independent international standard** and does not subordinate itself to any regional regulatory regime. Its interoperability strategy is "baseline compatibility, differentiated value":

- **Baseline compatibility (non-negotiable interop):** open data carriers, global identifiers (GTIN alongside DID), standard credential formats (W3C VC / SD-JWT VC), standard APIs, and machine-readable conformity claims. These ensure DPP-CQ credentials can be read and cross-checked by EU DPP, retail, customs and GS1-based systems.
- **Differentiated value (kept independent):** the quality-and-culture assessment methodology (Std 3001), the multi-stakeholder governance model, affordability for small producers, and applicability to cultural product categories that environmental DPPs do not cover.
- **Multilateral bridge:** active engagement with the UN/CEFACT UNTP (UN Transparency Protocol) conformity-credential vocabulary and the ISO DPP work item (ISO/AWI 25534-1 track), so that DPP-CQ interops through neutral, multilateral standards rather than through any single jurisdiction's registry.

DPP-CQ's flagship categories (tea, wine & spirits, cultural crafts, traditional medicine and health products, specialty agri-foods) are not within the first ESPR priority product groups; the standard therefore has a strategic window to mature its interop profile before any statutory DPP obligations potentially reach its categories.

8.3 Data Lifecycle Governance

Every DPP-CQ credential in v2.0 carries a `dataLifecycle` record identifying the data controller, retention policy, storage jurisdiction, cross-border mode (default: hash-only), deletion/anonymization rules, privacy notice URI, and PIA/DPIA summary URI. A public PIA summary is maintained at docs/compliance/pia-summary.md, and conformity declarations against GB/T 47507-2026, the ESPR/EN framework, and data-protection regimes are maintained at docs/compliance/conformity-declarations.md.

8.4 Electronic Evidence and Legal Effect

ICO standards are themselves technical and industry standards, without legal force (consistent with the nature of international standards such as those from ISO). However, their design fully considers judicial evidence requirements:

- **Hash integrity proof**: Cryptography-based data integrity verification can serve as electronic evidence in most jurisdictions
- **Trusted timestamps**: Integration with authoritative timestamp services meets the temporal certainty requirements for electronic evidence
- **Complete evidence chain**: Issuance chains, change records, and revocation logs constitute a complete evidence chain
- **Third-party audits**: Root nodes and core services undergo regular independent third-party audits to enhance evidentiary credibility

Specific legal effect varies by jurisdiction. ICO encourages regional nodes to collaborate with local legal institutions to promote recognition of the evidentiary weight of DPP-CQ credentials in their respective jurisdictions.

8.5 Dispute Resolution Mechanism

Referencing UNCITRAL (United Nations Commission on International Trade Law) arbitration rules and international commercial arbitration practices, ICO establishes a three-tier dispute resolution mechanism:

1. **Self-Help Mediation**: Parties negotiate independently through platform tools
2. **Expert Review**: A review panel composed of industry and technical experts issues a ruling
3. **Arbitration (Final)**: Submission to international commercial arbitration, with arbitral awards enforceable globally under the New York Convention

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9. Getting Involved

9.1 Membership Categories

Membership Category	Core Rights	Primary Obligations	Eligible Applicants
Founding Member	Council seats, priority participation, standard proposal rights	Annual contribution commitment, governance participation	Core founding institutions
Contributing Member	Working group participation, standard proposal rights, voting rights	Abide by charter, contribute to standard content	Enterprises, NGOs, academic institutions
Adopting Member	Free standard use, technical support, listing in ecosystem directory	Logo usage compliance, feedback on usage	Product manufacturers, service providers
Community Member	Participate in open-source community, submit feedback, attend public meetings	No mandatory obligations	Developers, researchers, individuals

9.2 Participation Pathways

Technical Participation: Join the GitHub open-source community and contribute code, documentation, and technical solutions

Standards Participation: Apply to join relevant standards Working Groups (WGs) to participate in standard development and review

Ecosystem Participation: Become a verified service provider, regional node operator, or joint laboratory partner

Adoption Participation: Implement ICO standards in products or services and apply for DPP-CQ conformance assessment

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10. Roadmap & Pilots

10.1 Three-Year Development Roadmap

2026: Foundation and Ecosystem Launch Year - Q3: Publish governance standards and DPP-CQ v1.0 - Q3: Establish Technical Secretariat, launch Root Resolver Network - Q4: Host Global Digital Credibility Governance Summit - Q4: Publish GDCC standard draft - Year-end target: 3+ standards published, 50+ member institutions, 1,000+ pilot products

2027: Ecosystem Expansion and Regional Deployment Year - Publish 2–3 new standards (GI mutual recognition, haute couture, etc.) - Launch 3 regional nodes: Southeast Asia, Middle East, Africa - Establish 5+ industry joint laboratories - 500+ institutional adopters, 10,000+ verified products - Establish formal cooperation with 2–3 international standards organizations

2028 and Beyond: Global Mutual Recognition and Ecosystem Maturity Year - 10+ standards published - 20+ national/regional nodes - 1,000+ institutional adopters, 100,000+ verified products - Establish mutual recognition mechanisms with major international standards organizations

10.2 Initial Pilot Framework

The first batch of flagship product assessment pilots will launch in the second half of 2026:

Pilot Category	Flagship Direction	Collaboration Model	Success Metrics
GI Tea	Top-tier production region GI tea	Production region + brand joint assessment	Number of verified SKUs, traceability data coverage rate
ICH Ceramics	Handcrafted technique heritage	Artisan + production region joint	Cultural data completeness, user verification volume
Haute Couture Attire	Chinese haute couture brands	Brand assessment + craft evidence anchoring	Full lifecycle passport coverage, market feedback
ICH Handicrafts	Embroidery / lacquerware, etc.	Transmitters + institutional joint	ICH value display, cross-border transaction improvement

Pilot evaluation dimensions: number of verified products, traceability data quality, user verification frequency, cross-border transaction improvement, producer value gain.

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11. Risks & Challenges

Risk	Description	Mitigation Strategy
Adoption Resistance	SMEs have limited digital capabilities and concerns about integration costs	Provide lightweight toolkits, SaaS platforms, capacity building programs; free during the first 6-month pilot period
Privacy and Data Security	High protection requirements for cultural craftsmanship and commercial data	Strict selective-disclosure application, off-chain storage architecture, regular third-party security audits; data control rests with issuers
Standards Fragmentation	Multiple sets of DPP standards coexist globally, creating confusion for enterprises	Actively promote mutual recognition and compatibility; participate in multilateral interoperability initiatives such as UNECE; avoid closed systems
Governance Neutrality	International community has concerns about the fairness of standard-setting	Transparent governance processes, multi-party oversight mechanisms, independent third-party audits, balanced representation
Technology Evolution Risk	Cryptography and blockchain technologies evolve rapidly, risking standard obsolescence	Modular design, extensible architecture, periodic technical reviews, backward compatibility strategy
Geopolitical Risk	Changes in international landscape may affect global acceptance of the standard	Maintain technology-neutral positioning, de-emphasize geopolitical coloring, focus on solving real problems as the core driving force

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12. Conclusion

In the digital age, trust is the most precious public good.

From the Industrial Revolution to the Information Revolution, every leap in human society has been accompanied by iterations in trust mechanisms. Today, as digital technology reshapes global trade and cultural exchange at an unprecedented pace, we need a more inclusive and pluralistic digital trust infrastructure — one that not only serves efficiency optimization for industrial products but also safeguards the diversity of human culture, ensuring that the unique value of every civilization can be seen, recognized, and respected in the digital world.

The ICO standards framework is committed to providing open, neutral, and inclusive digital trust infrastructure for global cultural and quality assets. We are not building a closed standards system; rather, we are contributing to the global digital trust edifice — using W3C's technical language to write chapters of cultural value; using multi-stakeholder governance wisdom to build cross-cultural bridges of trust.

We look forward to a future where every product carrying human wisdom and cultural value possesses a trustworthy digital identity — making global trade fairer, cultural heritage more vibrant, and the digital economy more human-centered.

We invite stakeholders worldwide to submit revision proposals by August 20, 2026 (Accelerated Procedure · 45-day review period), and to join us in building a more inclusive global digital trust future.

Annex A (Normative): DPP-CQ Technical Specification Draft

Annex A (Normative): DPP-CQ Technical Specification Draft

> This annex presents a summary of the DPP-CQ standard technical specification draft. The complete technical specification will be published in the official version of ICO Std 2001-2026.

A.1 DID Identifier Format

The DPP-CQ product DID uses the `did:ico:dpp` method name, with the following format:

```
did:ico:dpp:<namespace>:<unique-id>
```

- `namespace` : Category or issuer namespace (e.g., `tea` , `ceramic` , `fashion`)
- `unique-id` : Product unique identifier; UUID v4 or GS1-like coding is recommended

Examples:

```
did:ico:dpp:tea:longjing-xh-2026-001
did:ico:dpp:ceramic:jz-yd-2026-0892
```

A.2 JSON-LD Context Example

The JSON-LD context definition for DPP-CQ credentials:

```json
{
  "@context": [
    "https://www.w3.org/2018/credentials/v1",
    "https://www.w3.org/2018/credentials/examples/v1",
    "https://icoun.org/contexts/dpp-cq/v1"
  ],
  "id": "did:ico:dpp:tea:longjing-xh-2026-001",
  "type": ["VerifiableCredential", "DPPQualityCredential", "CulturalHeritageCredential"],
  "issuer": "did:ico:issuer:zjtea-assoc",
  "issuanceDate": "2026-07-15T00:00:00Z",
  "expirationDate": "2027-07-14T23:59:59Z",
  "credentialSubject": {
    "id": "did:ico:dpp:tea:longjing-xh-2026-001",
    "productName": "West Lake Longjing Tea",
    "category": "geographical-indication-tea",
    "qualityGrade": "Premium",
    "origin": {
      "type": "GeographicalIndication",
      "giNumber": "GI-CN-0001",
      "region": "West Lake, Hangzhou Production Area",
      "country": "CN"
    },
    "culturalData": {
      "heritageStatus": "National Intangible Cultural Heritage",
      "heritageYear": 2008,
      "craftMethod": "Hand-pan-fired · Ten Traditional Techniques",
      "artisanInfo": {
        "name": "(Optional, disclosed upon authorization)",
        "generation": "5th Generation Inheritor"
      }
    },
    "traceability": {
      "harvestDate": "2026-03-28",
      "processLocation": "Longwu Tea Town, Hangzhou",
      "batchNumber": "XH20260328A01"
    },
    "qualityAttributes": {
      "appearance": "Flat and smooth, tender green color",
      "aroma": "Lasting fresh aroma",
      "taste": "Fresh, mellow, and smooth"
    }
  },
  "proof": {
    "type": "BbsBlsSignature2020",
    "created": "2026-07-15T10:30:00Z",
    "proofPurpose": "assertionMethod",
    "verificationMethod": "did:ico:issuer:zjtea-assoc#keys-1",
    "proofValue": "..."
  }
}
```

A.3 Root Resolver Node API Specification (Draft)

Endpoint	Method	Function
`/v1/did/{did}`	GET	Query DID document
`/v1/credential/{credentialId}/status`	GET	Query credential status (valid/revoked/expired)
`/v1/credential/verify`	POST	Verify credential signature and validity
`/v1/register`	POST	Register new credential hash
`/v1/revoke`	POST	Revoke credential
`/v1/proof/{hash}`	GET	Obtain hash anchoring proof

A.4 Three-Tier Verification Process

```
Scan QR / NFC Tap
     │
     ├── Quick Verification (<1s)
     │   └── Verifies: DID validity + Credential status + Basic information
     │
     ├── Deep Verification (3–5s)
     │   └── Verifies: Complete credential signature chain + Traceability data
     │                    + Quality details + ZKP proof
     │
     └── Judicial-Grade Verification (application required)
         └── Verifies: Root node anchored hash + Timestamp proof
                  + Complete audit trail
```

Annex B (Informative): Glossary

Annex B (Informative): Glossary of Terms

Term	Definition
Digital Product Passport (DPP)	A digital identity credential containing core product data, readable via methods such as code scanning
Decentralized Identifier (DID)	A decentralized identity identifier defined by W3C, independent of centralized identity providers
Verifiable Credential (VC)	A digital credential digitally signed by the issuer, independently verifiable by third parties
Selective Disclosure (SD-JWT / BBS+)	Cryptographic proofs (SD-JWT per RFC 9529; BBS+ unlinkable derived credentials) that let a holder reveal only selected claims while keeping the signature valid; v1.x used the umbrella term "zero-knowledge proofs (ZKP)", v2.0 normatively uses these standard suites
Selective Disclosure	The practice by which a data holder discloses only the necessary data fields to a verifier
Geographical Indication (GI)	An indication identifying a product as originating from a specific territory, where a given quality, reputation, or other characteristic of the product is essentially attributable to its geographical origin
Intangible Cultural Heritage (ICH)	Practices and expressions recognized by communities, groups, and, in some cases, individuals as part of their cultural heritage, as defined by UNESCO
Root Resolver	A unified resolution and anchoring infrastructure for globally distributed identifiers
Hash Anchoring	Storing hash fingerprints of data on tamper-proof media for integrity verification and timestamping
JSON-LD	JSON for Linked Data — a JSON format for linked data, supporting semantic interoperability
Multi-Stakeholder Governance	A governance model ensuring that all parties affected by decisions have the opportunity to participate in decision-making

Annex C (Informative): Abbreviations

Annex C (Informative): Abbreviations

Abbreviation	Full Form
ICO	International Communication Organization
DPP	Digital Product Passport
DPP-CQ	Digital Product Passport for Cultural & Quality Goods
DID	Decentralized Identifier
VC	Verifiable Credential
ZKP	Zero-Knowledge Proof
GI	Geographical Indication
ICH	Intangible Cultural Heritage
GDCC	Global Digital Credibility Assessment
TCR	Tianji Credibility Registry
GDC	Global Digital Compact
W3C	World Wide Web Consortium
WIPO	World Intellectual Property Organization
UNESCO	United Nations Educational, Scientific and Cultural Organization
ISO	International Organization for Standardization
IETF	Internet Engineering Task Force
UNECE	United Nations Economic Commission for Europe
ESPR	Ecodesign for Sustainable Products Regulation
GS1	Global Standard 1
GDPR	General Data Protection Regulation
UNCITRAL	United Nations Commission on International Trade Law
NFC	Near Field Communication
CLA	Contributor License Agreement

Annex D (Informative): References

Annex D (Informative): References

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Annex E (Informative): Frequently Asked Questions

Annex E (Informative): Frequently Asked Questions

Q1: What is ICO? What qualifications does it have to develop international standards?

ICO (International Communication Organization) is a multi-stakeholder international cooperation platform dedicated to advancing global digital inclusion and cross-cultural digital trust.

The ICO standards framework employs an open community-driven development model, following best practices in international standard-setting (openness, transparency, consensus, balanced representation). The vitality of standards lies in adoption and ecosystem, not administrative mandate — the core Internet protocols (TCP/IP, HTTP, etc.) were not promulgated by governments or official bodies, but became de facto standards through joint advancement by open-source communities and standards organizations and widespread market adoption. ICO is committed to becoming the de facto standard-setter in the field of cultural and quality digital trust.

Q2: What is the relationship between ICO standards and the EU DPP? Will they conflict?

A complementary relationship, not a competitive one.

The EU DPP (under the ESPR framework) focuses primarily on environmental and circular economy data for industrial products — a "green dimension" Digital Product Passport. ICO DPP-CQ focuses on the cultural value and quality dimensions of cultural products and specialty agricultural goods — a "cultural dimension" Digital Product Passport. The two cover different product categories and data dimensions; they are complementary rather than substitutive.

At the technical level, DPP-CQ is fully compatible with W3C DID/VC standards, sharing the same technical foundation as the EU DPP, enabling data interoperability. v2.0 makes this concrete: products in the GS1 interop profile carry GTIN identifiers and open GS1 Digital Link carriers (aligned with EN 18219/18220), credentials support SD-JWT VC (RFC 9529) alongside BBS+, and sustainability data follows ISO 14067 and UNTP-style conformity claims, so EU importers and regulators can read and cross-check DPP-CQ data with standard tooling. DPP-CQ's flagship categories (tea, wine & spirits, cultural crafts, traditional medicine products) are not in the first ESPR priority product groups, so there is no statutory conflict; the two regimes address different dimensions and can be carried by the same passport. We welcome standard coordination and mutual recognition dialogue with relevant EU institutions.

Q3: Why is a Root Resolver Network needed? Will it create a monopoly?

The core function of the Root Resolver Network is to ensure consistency of DID resolution and queryability of credential status in global cross-border circulation. This is analogous to the root servers of the Domain Name System (DNS) — not for the purpose of monopolization, but for the purpose of interoperability.

The design of the root resolution mechanism follows the following principles: **- Technology-neutral**: Root nodes only provide resolution and anchoring services; they do not participate in standard-setting and do not engage in commercial competition; **- Governance constraints**: Root node operation is overseen by the Technical Committee and authorized by the Standards Council; major changes require approval through governance processes; **- Gradual decentralization**: The long-term goal is to establish a distributed root node network with multiple regions and multiple operators, avoiding single-point dependencies; **- Open source and open**: The reference implementation of root nodes is fully open-source, and any institution may deploy a compatible node.

Q4: How are data security and privacy ensured? Will data exit issues arise?

ICO standards employ an architectural design of "data localization, hash cross-border": **- Raw data stored locally**: Detailed product quality, craftsmanship, and supply chain data are stored locally by the issuer or certification body, with no mandatory cross-border transmission; **- Hash fingerprint anchoring**: Only hash fingerprints of data (one-way irreversible) are registered and anchored at root nodes; hashes themselves contain no identifiable original information; **- Selective disclosure**: Sensitive data is verified via SD-JWT / BBS+ proofs; verifiers receive only the claims or compliance conclusions they are authorized for, without access to undisclosed raw data; **- Compliance adaptation**: Regional nodes may conduct localized deployment and compliance adjustments in accordance with local data regulations.

This architecture fundamentally reduces the risk of cross-border flow of raw data and complies with regulatory requirements for data sovereignty and data security in all countries.

Q5: What is the relationship between the Tianji system and ICO?

The Tianji system is one of the institutions undertaking the role of Technical Secretariat for the ICO standards framework, responsible for reference implementation of standards, root node operation and maintenance, and technical support.

This arrangement is analogous to the relationship between W3C and host institutions such as MIT/ERCIM — standards are developed jointly by the community, and the Technical Secretariat is responsible for execution and implementation. The Tianji system, as a technical service provider, does not participate in standard voting, ensuring separation of standard-setting from technical execution.

Q6: Is there a fee to adopt ICO standards?

The standards themselves are completely free. - Standards documentation: Free to download and use (CC BY 4.0 / Apache 2.0) - Reference implementations: Open-source and free, freely modifiable and commercially usable - Basic verification: Basic functions such as scan-to-verify are freely available

Value-added services (such as custom development, advanced assessment services, technical support, etc.) may be provided by service providers in the ecosystem; ICO itself does not directly operate commercial services.

Q7: What is the relationship with GS1 and ISO? Is this a separate stovepipe effort?

No, it is not a separate stovepipe effort. ICO standards are built upon existing international standards such as W3C, ISO, and GS1, serving as a supplement and extension of the existing standards ecosystem in the cultural and quality dimensions.

W3C DID is the "technical standard for ID cards" (telling us how to make the card), ISO/GS1 is the "industry coding specification" (telling us how to number things), and ICO DPP-CQ is the "content specification for cultural product passports" (telling us what content goes in the passport and how to verify cultural value). We use W3C's card technology, are compatible with GS1's coding system, and focus on content standards for the vertical domain of culture and quality.

We actively seek cooperation and mutual recognition with standards organizations such as ISO, W3C, and GS1.

Q8: Can small and medium-sized producers afford it? Will it increase their burden?

This is precisely one of the core concerns of ICO standards.

Our design goal is to enable small and medium-sized producers and ICH transmitters to "afford and benefit from" the system: - **Low-cost solutions**: Support low-cost verification methods such as QR codes, with startup costs as low as a few cents per item; - **Lightweight onboarding**: SaaS-based assessment platforms are provided, requiring no technical team — usable upon registration; - **Volume discounts**: Discounts or waivers for small and medium-sized producers and ICH transmitters; - **Capacity building**: Training and technical support provided through regional nodes and partners.

We believe that the "premium for quality" effect brought by digital passports will far exceed their usage cost — helping good products command good prices is the most direct value empowerment for producers.

Q9: What about the legality and evidentiary effect of the standards?

ICO standards are themselves a set of technical and industry standards, without legal force (consistent with the nature of international standards such as those from ISO). However, their design fully considers judicial evidence requirements:

- **Hash anchoring**: Cryptography-based data integrity proof can serve as electronic evidence in most jurisdictions;
- **Trusted timestamps**: Integration with authoritative timestamp services meets the temporal certainty requirements for electronic evidence;
- **Complete evidence chain**: Issuance chains, change records, and revocation logs constitute a complete evidence chain;
- **Third-party audits**: Root nodes and core services undergo regular independent third-party audits to enhance evidentiary credibility.

Specific legal effect varies by jurisdiction. ICO encourages regional nodes to collaborate with local legal institutions to promote recognition of the evidentiary weight of DPP-CQ credentials in their respective jurisdictions.

Q10: How is the neutrality of the standards ensured? How can we be certain no single party dominates?

The multi-stakeholder governance mechanism is the institutional guarantee for standard neutrality:

- **Balanced representation**: Council seats are allocated by category (government 25%, industry 25%, academia 20%, civil society 20%, Technical Secretariat 10% non-voting), so no single group can dominate decision-making;
- **Consensus-first**: Important decisions prioritize seeking consensus rather than simple majority voting;
- **Separation of technology and standards**: The Technical Secretariat does not participate in standard voting, separating execution from decision-making;
- **Transparency and openness**: Governance processes are open and transparent, subject to community oversight;
- **Open participation**: Any institution may apply to join, regardless of nationality, size, or ownership structure.

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📥 **Download PDF:** White Paper v2.0 (PDF, includes all normative annexes)

Annex F (Normative): v2.0 Interoperability & Compliance Annexes

Annex F (Normative): v2.0 Interoperability & Compliance Annexes

The following separate documents form a normative part of ICO Std 2001 v2.0. They are maintained in the repository under `docs/`:

Document	Path	Content
GS1 Digital Link & Identifier Mapping	`docs/interoperability/gs1-digital-link-mapping.md`	GTIN ↔ DID binding rules, Digital Link URI structure and content negotiation, EPCIS 2.0 event mapping, EN 18219/18220 alignment
Data Carrier Specification	`docs/interoperability/carrier-specification.md`	Three-tier carriers (L1 open QR / L2 SDM NFC / L3 tamper-evident NFC), open-carrier principle, dual-carrier rule, resolver behavior
Credential Format Profiles	`docs/specs/credential-formats.md`	JSON-LD Data Integrity (BBS+) vs SD-JWT VC (RFC 9529) profiles; disclosure classification; SM2/SM3/SM4 national cryptography suite; status and validity rules
Conformity Declarations	`docs/compliance/conformity-declarations.md`	Self-declarations against GB/T 47507-2026, EU ESPR/EN standards, GDPR/PIPL, EU AI Act
PIA / DPIA Summary	`docs/compliance/pia-summary.md`	Public privacy impact assessment: data flows, risk table, data-subject rights, cross-border legal basis
JSON Schema v2.0	`schemas/dpp-cq.schema.json`	Machine-readable data model ($id: icoun.org/schemas/dpp-cq/v2.0.0-draft)
v2.0 Example Credential	`examples/longjing-tea-v2.json`	Full interop-profile example (Longjing tea)

Relationship to external frameworks

External framework	Nature	DPP-CQ posture
W3C DID / VC v2.0, IETF SD-JWT (RFC 9529)	Open standards	Foundation; fully adopted
GS1 Digital Link / EPCIS 2.0	Global supply-chain standards	Normative interop profile
EU ESPR (Reg. 2024/1781); EN 18219–18223	EU regulation / harmonized standards	Baseline compatibility for carriers, identifiers, sustainability fields; DPP-CQ categories not in first priority groups
UN/CEFACT UNTP	UN multilateral vocabulary	Adopted for conformity claims; primary multilateral bridge
ISO 14067, ISO 22000, ISO 22739	ISO standards	Referenced in data model and terminology
GB/T 47507-2026; SM2/SM3/SM4 (GB/T 32918/32905/32907)	Chinese national standards	Supported for domestic deployments
EU AI Act (Reg. 2024/1689) Art. 50	EU regulation	Transparency provisions adopted as baseline practice

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